

MATH 2211, SPRING 2026: HOMEWORK 4

- (1) Decide if the following statements are true or false. Give a line or two in justification for your answer. It is permissible for that justification to be a reference to previous homeworks or to lectures:

- (a) The polynomials $1, x^2, i + ix^2$ in $\text{Poly}_{\leq 2}(\mathbb{C})$ are linearly independent.
- (b) Any set of vectors containing 0 cannot be linearly independent.
- (c) Every subspace of a vector space V is a span of some subset of V .
- (d) The intersection of two subspaces of a vector space V is once again a subspace of V .
- (e) If $\{v_1, v_2\}$ is a linearly independent subset of V and $v_3 \notin \text{span}(\{v_1, v_2\})$, then $\{v_1, v_2, v_3\}$ is a linearly independent subset as well.

- (2) Consider the subset

$$U = \{f(x) \in \text{Poly}_{\leq 3}(\mathbb{R}) : \int_0^1 f(x) dx = 0\} \subset \text{Poly}_{\leq 3}(\mathbb{R}).$$

- (a) Show that U is a subspace of $\text{Poly}_{\leq 3}(\mathbb{R})$.
- (b) Find a basis for U and extend it to a basis for $\text{Poly}_{\leq 3}(\mathbb{R})$. What is the dimension of U ?

- (3) Consider the subset

$$V = \{f(x) \in \text{Poly}_{\leq 5}(\mathbb{Q}) : f(1) = f(2) = 0\} \subset \text{Poly}_{\leq 5}(\mathbb{Q}).$$

- (a) Show that V is a subspace of $\text{Poly}_{\leq 5}(\mathbb{Q})$.
- (b) Find a basis for V and extend it to a basis for $\text{Poly}_{\leq 5}(\mathbb{Q})$. What is the dimension of V ?

- (4) Consider the subset

$$W = \left\{ A \in M_{2 \times 2}(\mathbb{Z}/3\mathbb{Z}) : \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} A = A \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \right\}.$$

- (a) Show that W is a subspace of $M_{2 \times 2}(\mathbb{Z}/3\mathbb{Z})$.
- (b) Find a basis for W and extend it to a basis for $M_{2 \times 2}(\mathbb{Z}/3\mathbb{Z})$. What is the dimension of W ?

- (5) If $f(x)$ is a polynomial of degree n over $\mathbb{R}$, show that the polynomials

$$1, f(x), f'(x), \dots, f^{(n-1)}(x)$$

form a basis for $\text{Poly}_{\leq n}(\mathbb{R})$.

- (6) Compute the change of basis matrices (in both directions) between the bases

$$\mathcal{B}_1 = \{1, x, x^2, x^3\}; \mathcal{B}_2 = \{1, x-2, (x-2)^2, (x-2)^3\}$$

for the $\mathbb{Z}/5\mathbb{Z}$ -vector space $\text{Poly}_{\leq 3}(\mathbb{Z}/5\mathbb{Z})$.